

Supplementary Material for: GroundSLAM: A Robust Visual SLAM System for Warehouse Robots Using Ground Textures

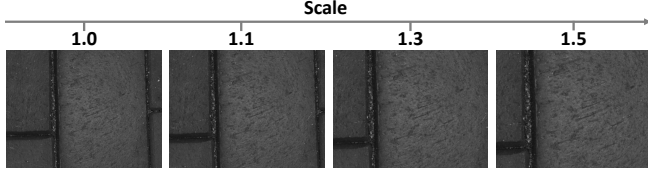


Fig. 1: We apply a scale transformation to one image in each image pair to simulate the effect of relative scale differences caused by changes in camera height. A demonstration video is available here: <https://tinyurl.com/488d5yt9>.

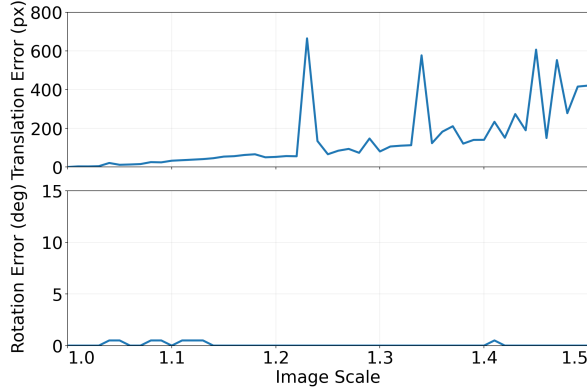


Fig. 2: This figure shows the translation and rotation errors produced by our method when estimating relative poses from image pairs with a resolution of 1600×1200 at different scales.

1. Additional Experimental Results

1.1 Robustness to Out-of-Plane Motion

To evaluate the robustness of GroundSLAM against out-of-plane motions, we conducted controlled experiments on the HD Ground dataset to simulate vertical displacement, pitch, and roll perturbations. These experiments quantitatively assess the sensitivity of the proposed system to each type of motion, as discussed below.

1) Vertical displacement. A change in camera height mainly appears as an image-scale variation. We therefore simulate vertical motion by applying a scaling factor s to one image of each pair. Since the problem depends only on the relative scale between the two images, scaling one image by s is equivalent to scaling the other by $1/s$. To avoid invalid regions caused by image reduction, we only enlarge one image, with s varying from 1.0 to 1.5 in increments of 0.01. The camera height in the HD Ground dataset is 23.4 cm; under the pinhole-camera approximation,

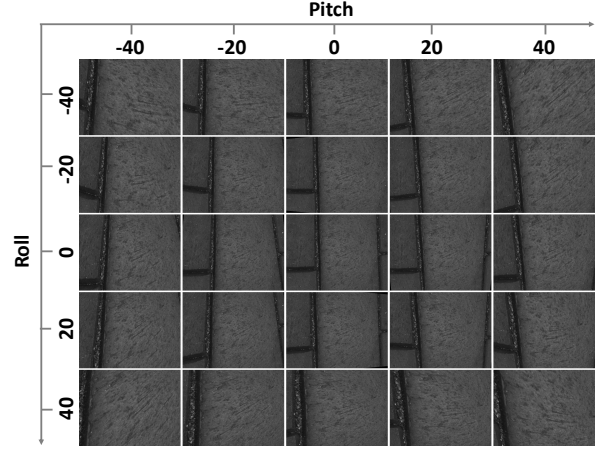


Fig. 3: We apply a homography transformation to one image in each image pair to simulate the effect of pitch and roll attitude changes. A demonstration video is available here: <https://tinyurl.com/488d5yt9>.

the maximum scale factor of 1.5 corresponds to a height reduction of approximately 7.8 cm. Fig. 1 shows an example of an image under different scale transformations, while Fig. 2 shows the translation and rotation pose estimation errors at different relative scales.

The results show that rotation estimation is only mildly affected by scale variation. The translation error also remains relatively small when the relative scale is below 1.2. A scale factor of 1.2 corresponds to a height variation of approximately 16.7%, or about 4 cm for the evaluated camera setup. These results indicate that GroundSLAM has a certain tolerance to the moderate vertical displacement.

2) Pitch and Roll attitude changes. To evaluate the robustness of GroundSLAM against small camera attitude variations, we simulate roll and pitch perturbations by applying a projective transformation to the second image of each image pair. Let α and β denote the roll and pitch perturbations, respectively. The corresponding rotation matrices are

$$\begin{aligned} \mathbf{R}_x(\alpha) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}, \\ \mathbf{R}_y(\beta) &= \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}. \end{aligned} \quad (1)$$

The combined rotation is defined as $\mathbf{R}(\alpha, \beta) = \mathbf{R}_y(\beta)\mathbf{R}_x(\alpha)$ which corresponds to applying roll first and

TABLE I: ATE RMSE of different systems on the real-world rough-surface textures (unit: m).

Texture	6-DOF Pose Estimation		3-DOF Pose Estimation	
	ORB-SLAM3 [1]	SVO [2]	GT-SLAM [3]	Ours
Gravel Road	0.320	<u>0.291</u>	0.772	0.253
Brick	0.177	0.244	0.684	<u>0.188</u>
Asphalt	<u>0.120</u>	0.082	0.217	0.163

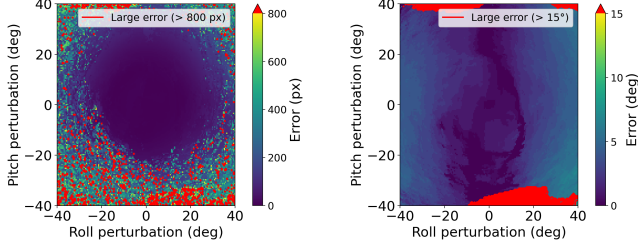


Fig. 4: This figure shows the translation and rotation errors produced by our method when estimating relative poses from 1600×1200 image pairs under pitch and roll variations.

pitch second. Given the calibrated camera intrinsic matrix $\mathbf{K}$, the image homography induced by this camera rotation is then computed as

$$\mathbf{H} = \mathbf{K}\mathbf{R}(\alpha, \beta)\mathbf{K}^{-1}. \quad (2)$$

A direct application of $\mathbf{H}$ may introduce a global displacement of the image center. To isolate the projective deformation caused by roll and pitch from this global image translation, we constrain the geometric image center to remain fixed. The center is represented in homogeneous coordinates as

$$\mathbf{c}^{\text{img}} = [(W-1)/2 \quad (H-1)/2 \quad 1]^T, \quad (3)$$

where W and H are the image width and height. Its position after the rotation-induced homography is

$$\mathbf{c}' \sim \mathbf{H}\mathbf{c}, \quad (4)$$

where $\sim$ denotes equality up to a non-zero scale factor. After dehomogenization, the translation required to restore the original image center is

$$\Delta \mathbf{c} = \begin{bmatrix} c_x^{\text{img}} - c'_x \\ c_y^{\text{img}} - c'_y \end{bmatrix}. \quad (5)$$

The corresponding compensation matrix is

$$\mathbf{T}_c = \begin{bmatrix} 1 & 0 & c_x^{\text{img}} - c'_x \\ 0 & 1 & c_y^{\text{img}} - c'_y \\ 0 & 0 & 1 \end{bmatrix}. \quad (6)$$

The final transformation used to generate the perturbed image is therefore

$$\mathbf{H}_{\text{final}} = \mathbf{T}_c \mathbf{K} \mathbf{R}_y(\beta) \mathbf{R}_x(\alpha) \mathbf{K}^{-1}. \quad (7)$$

The transformed image is generated using perspective warping while preserving the original image resolution. Pixels outside the valid source image are filled with zero. A joint roll-pitch perturbation is obtained by evaluating a two-dimensional grid of (α, β) values, and both angles are varied

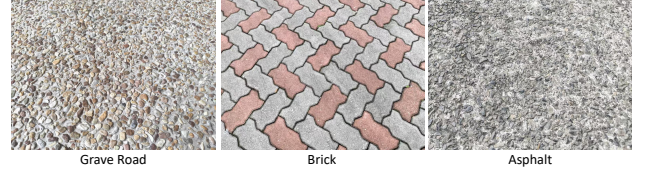


Fig. 5: Examples of the rough ground surfaces used to evaluate robustness against out-of-plane disturbances.

from -40° to 40° with an interval of 0.5° . Fig. 3 shows an example image under different pitch and roll transformations, while Fig. 4 presents the translation and rotation errors of our method under varying pitch and roll transformations.

The results show that both translation and rotation errors remain relatively low when the pitch and roll angles are below 20° , indicating that the proposed method has a certain tolerance to these two types of out-of-plane rotation.

3) Experiments on Real-World Data. We also collected real-world sequences on rough surfaces containing large cracks, loose gravel, and other irregularities that induce robot vibrations, vertical displacement, and pitch and roll perturbations. Example images of these surfaces are shown in Fig. 5. The trajectory errors of ORB-SLAM, SVO, GT-SLAM, and GroundSLAM are reported in Table I.

The results show that GroundSLAM maintains high localization accuracy on these rough surfaces and achieves performance comparable to that of systems estimating full 6-DOF motion. This further demonstrates the robustness of the proposed method to moderate out-of-plane disturbances.

4) Conclusion. Overall, the three experiments indicate that GroundSLAM can tolerate mild out-of-plane motion. When the out-of-plane motion is limited, it introduces some pose estimation errors, but these errors remain relatively small. However, the estimation error increases significantly once the disturbance exceeds a certain range. In our target application, modern warehouse floors are generally relatively flat, and we have not observed such motions to noticeably affect localization performance during practical deployment.

References

- [1] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. Montiel, and J. D. Tardós, “Orb-slam3: An accurate open-source library for visual, visual-inertial, and multimap slam,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.
- [2] C. Forster, Z. Zhang, M. Gassner, M. Werlberger, and D. Scaramuzza, “SVO: Semidirect visual odometry for monocular and multicamera systems,” *IEEE Transactions on Robotics*, vol. 33, no. 2, pp. 249–265, 2017.
- [3] K. M. Hart, B. Englot, R. P. O’Shea, J. D. Kelly, and D. Martinez, “Monocular simultaneous localization and mapping using ground textures,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 2032–2038.